Book Bot: Exploring the Digital Frontier of Library Management through Automated Programs

A System Presented to the

Faculty of College of Arts and Sciences

Rizal Technological University

In Partial Fulfilment of the Requirements

for the Degree of Bachelor of Science in Statistics

by

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Description of the Proposed System

Libraries are an essential part of society and, most importantly, of any learning institution. They offer a wide variety of information and knowledge to the general public, and with the increase of digital media, traditional libraries are having a difficult time adjusting to the rapid change in society’s transition to digital advancements. This automated inventory system allows users to access, easily locate, and find the exact location of any resource material needed on one screen.

Current scholars at Rizal Technological University undergo numerous research papers, critical essays, and reflection papers on different and certain topics that require extensive and proper information. The College of Arts and Sciences (CAS) library offers a variety of different types of information. However, with that many resources, it can be daunting for every student to know where to start looking for the right information; even the management that handles the library’s archives can find it difficult to locate and organize every book available. The system proposes the use of the database from the College of Arts and Sciences thesis inventory, which will be provided in an external Excel file converted to a CSV. The program created can be used offline to easily locate any thesis within the library’s collection—from title, author, focus (course), year, location, and call number. This allows students, faculty, and, most importantly, the librarian to easily locate and organize the library's thesis archives with and without an internet connection.

As the program starts, the user will be asked to press enter to continue and enter the keywords they want to search. After the user enters the keyword or keywords, the program will inspect if the entered keyword or keywords are valid and available in the database. The program will display the "Thesis Code," "Focus" or college program, and "Title" of theses that contain keywords that the user entered. Then the program will ask the user to enter the code for the respective thesis they are looking for and display the rest of the information, like the date of publication, call number, and its location in the library. At the end of the program, it will ask the user if they want to repeat the search. If they enter "yes", the program will start again from scratch. And if they enter "no", the program will end.

Purpose

The automated inventory system provides a ton of benefits for the College of Arts and Sciences student body and faculty, especially for the library that houses multiple collections of the theses created by past scholars for this generation of scholars. The program took into consideration the need for organization and visibility in regard to creating this program. Since the objective of any automated program is to facilitate easier integration to carry out certain daily tasks, this program aims to provide convenience and easier data management and is accessible offline without using the internet that the current institution provides.

To improve organization and cataloguing of the library's archives.

The College of Arts and Sciences library contains most of the past scholars' academic works, such as different studies, research papers, and theses. Collectively, these can amount to a large collection over the years that can be difficult to maintain for easier access for the students and, most importantly, the librarian. These academic resources are beneficial for current students, especially as resources for conducting different academic papers and research studies. That is why this program was created to help provide the user's access to cumulatively improve the overall system of the library by providing one-stop access to the entire archive of the library—easier storage, management, and retrieval of records.

Offer an enhanced user experience.

Automated library management can help increase the user experience for students and faculty. By efficiently executing a solution, they can easily navigate through the struggles of trying to find their needed resources from countless mountainous books. The program created can be accessed offline compared to the already existing system, which requires the internet to navigate through the collection of books. The program itself can reduce the workload of the school's library management while increasing overall user satisfaction and experience.

Provide better accessibility to the archives.

With a large selection of various theses with different topics, it can be tedious and time-consuming to find the exact information needed, and research and researchers find how important time is for every college student. Integrating an automated system into the university's library can help students, faculty, and, most importantly, the library management easily access specific information associated with their curriculum, either by books, journals, articles, or—in this case—theses.

To conclude, the main objective of this program is to provide a one-stop access or compiler of the available theses in the school library archives that is easily accessible by the user to locate and help them find the source they require accurately for their academic needs.

Scope

This automated inventory system focuses on providing its users with a program that will help them easily locate thesis books within the library. The program also provides the means for easier management and organization of the thesis inventory. For this program, only ten (10) theses done during the year 2022 from each course in the College of Arts and Science will be used for its database, for a total of fifty (50) theses. To demonstrate the scope of this system, the programmers will enumerate the limitations on access and functions that the program can provide its users and administrators.

User

1. These are ordinary users of the program, may it be students or faculty, whose access is limited to searching for the theses they need to locate within the CAS Library.

2. The system will only provide the following information regarding a thesis book searched by its user: 1) title, 2) author, 3) focus, 4) year, 5) location, and 6) call number.
3. The keywords that the user will input in the search bar will only yield results from similar words found within the data in the CSV file.

Admin

1. This is the librarian, whose duty it is to manage and organize the thesis books within the library. Their primary role is to oversee and manage the database within the program, but they also have the same access as ordinary users.
2. The database will solely be provided in an external CSV file that can be managed through Excel or other similar spreadsheet software programs. This is to streamline the system's code, wherein all the needed data is within the specified CSV file and is simply imported into the program.
3. For further data to be added to the database, the admin will have to manually input this data by editing the CSV file in Excel or any similar spreadsheet software program.

In addition, external factors such as disorganized bookshelves and misplaced books that may impede in locating theses searched within the program are solely the responsibility of the library itself. The program only provides the necessary information that would help locate these theses within the library; the organization of these books still falls under the management of the library. The program also operates under existing rules and systems within the library.

Definition of Terms

Accessibility - refers to the availability and ease of use of services or information.

Automated - using technology and programs to perform daily tasks with minimal human intervention.

Cataloguing - creating an organized list or inventory of objects in a collection, such as books or research papers.

Compiler - a tool or program that gathers and organizes data from multiple sources into a cohesive structure, like a database.

CSV - (Comma-Separated Values): A file format for tabular data stored in plain text, commonly used for spreadsheets.

Csv_reader - an object created using the "csv.DictReader" function, allowing iteration over CSV file rows as dictionaries.

Database - an organized collection of data in a computer system designed for easy access and management.

Date Created - a column name in the CSV file for the dates of the books.

DictReader - a class in the CSV module used to read CSV files as dictionaries.

File_path - a variable storing the path of the CSV file to be processed by the code.

For loop - a control flow statement that iterates through a list of elements, running a code block for each.

Found_cells - an empty list in the code storing cells (column and value) matching specified keywords during the search.

f"*" * 100 - an f-string that creates a string of 100 asterisks.

F-string - a string literal in Python that allows for embedding expressions inside string literals, using curly braces {}.

if row['CODE'] in selected_codes - a conditional statement that checks if the value of the 'CODE' key in the current row is in the selected_codes list.

if statement - a conditional statement in the code checking if the "CODE" column value in the current row matches selected codes.

If statement[2] - a control flow statement that runs a code block if a predetermined condition is met.

Integration - the process of combining disparate components or systems into a cohesive whole.

Inventory - a comprehensive list or documentation of goods or resources in a specific location.

open() - a built-in function in Python used to open a file and return a file object.

print() - is a built-in Python function that shows output on the console.

Print_selected_rows - a function in the code that prints rows in a CSV file matching selected codes.

Repeat - a variable storing the user's response on whether to repeat the search, used to control the program's execution in a loop.

Retrieval - locating and gaining access to data or resources within a database or collection.

Row - a variable representing a single row in the CSV file, obtained by iterating over the "csv_reader" object.

row['Title'] - accessing the value of the 'Title' key in the current row dictionary.

Row² - a variable in a CSV file that represents a row; a dictionary with the cell values for that row as the values and the column names as the keys.

Search_and_list_cells - a function in the code that searches and lists cells in a CSV file matching specified keywords.

Selected_codes - a list of book codes chosen for printing.

shutil.get_terminal_size().columns - retrieves the number of columns in the current terminal window using the shutil module in Python. It allows you to dynamically adjust formatting or layout based on the available space in the terminal.

Thesis - an official academic paper summarizing a researcher's ideas and findings on a chosen topic.

Time - a Python module used to *import* provides functions and methods for working with time-related operations such as retrieving the current time, pausing execution, formatting time values, and measuring time intervals.

With - a context manager in Python used to simplify exception handling and resource management.

'R' - a mode parameter used in the open() function to open a file for reading.

\n: A unique character in Python to represent a new line.

Shutil - shutil or *shell utilities* is a high-level interface to file and directory operations Python module. It has tools to copy, move, delete, and archive files and directories. You may import "shutil" to execute the usual file operations and access these functions.

Systems Available in the Global Market

Several *automated management systems* are already available on the market. Every learning institution has a library, which is crucial for every student and faculty as the library provides the resources and information needed for various academic purposes. Although most libraries do not need to be automated, it does help the librarian maintain an inventory of both issued and available books. *It* was made to manage each part of the library's history, assisting librarians in keeping the database up-to-date.

Nowadays, the library system in the institution is much more unorganized since many undergraduate students’ degrees have their major papers placed on the shelf, and often researchers' major papers are just left on the floor. Poorly managed libraries have issues like disordered shelves and out-of-date references and are incapable of accommodating many papers; they proposed an automated inventory system to make things easier. It will work like any other bot available in applications like the Discord app, Messenger, and ChatGPT.

The program focuses more on students’ thesis papers with a user-friendly feature database for our scholarly literature that can help librarians manage thesis papers by storing all information such as thesis title, author name, rack detail, and keywords or abstracts. It provides students and teachers with a way to search for any thesis papers with a smooth experience using a single interface and ensures that it does not require technical expertise. Also, it efficiently keeps numerous theses collection trackers from our previous scholars. By increasing information and access for current and future scholars, they will make this program more comprehensive and user-centered.

Enhancement

As the digital landscape continues to develop over time to cope with the evolving demands of a tech-driven society, the same goal is expected for this automated inventory system. To create a more efficient automated inventory system that serves the students and faculty of RTU, the programmers recommend the following enhancements:

Internal Database

The program relies on an external Excel file converted to CSV as its database. To streamline the coding process, this CSV file is simply imported within the program. Future development for this program can explore having the database within the program itself. In this case, the system can be self-sufficient.

Login System

The program is readily accessible to anyone who opens it, both users and administrators. With an external database, the admin will have sole control over its management to further input and/or organize the data within the database. If the database is within the program itself, the program will need a login system to separate the functions that an ordinary user and an admin can access.

Data

The program used only ten (10) theses done during the year 2022 from each course within the College of Arts and Science for its database. Future development for this program can explore creating a wider database that includes theses from other libraries, made from any year, any college, and any course within the university. Books within the library other than theses can also be added to the database.

Book Tracking

Automation of the book borrowing process can also be explored for this program. This feature will give the admin the ability to digitally track the books borrowed by students and faculty, and these users will be able to digitally log the books they borrow. Additional features within this can include necessary book borrowing information such as informing the user of when their chosen book/s will be due upon logging them in for borrowing.

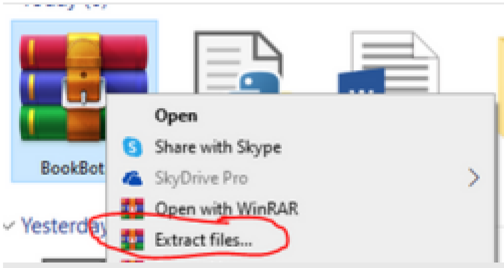
User Manual

Installation:

1. Download the folder from the provided link

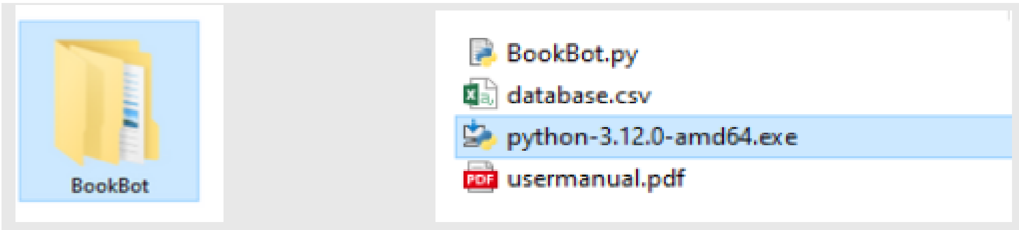
    https://rb.gy/si5anz

2. Extract the files to the desired location



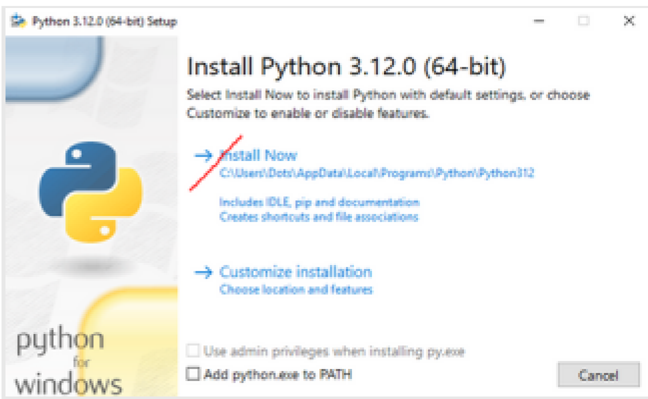
**Normally, it will redirect to 'C:\Downloads\BookBot

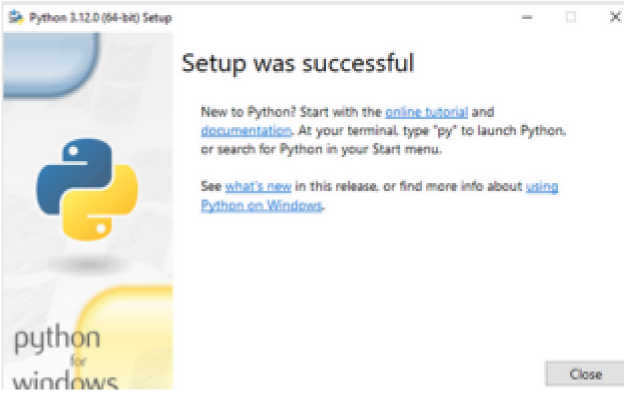
3. Open the extracted 'BookBot' folder.



4. Install 'Python-3.12.0-amd64.exe

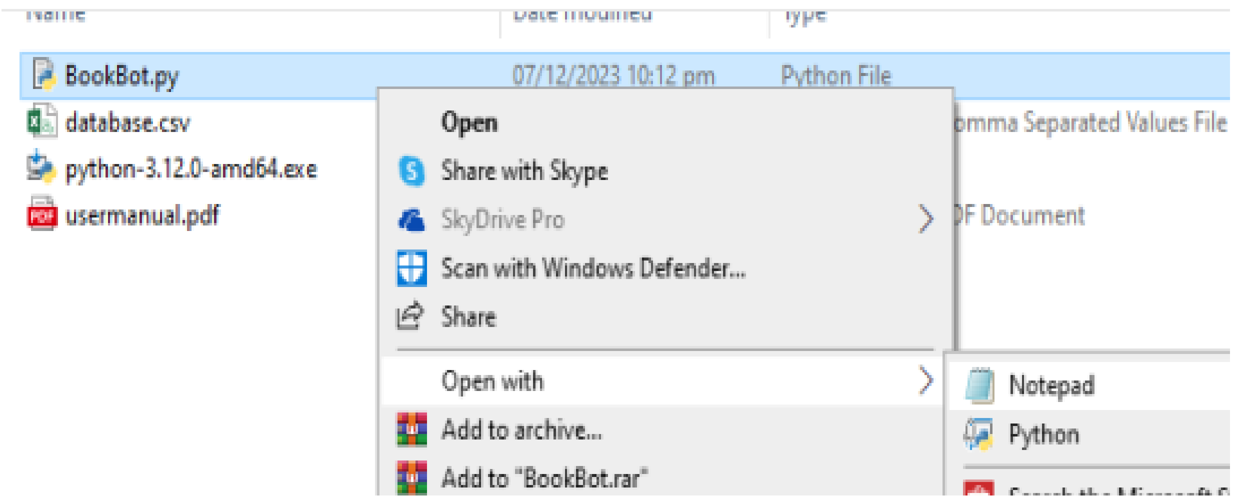
    **You can skip this step if you already have Python on your Desktop.



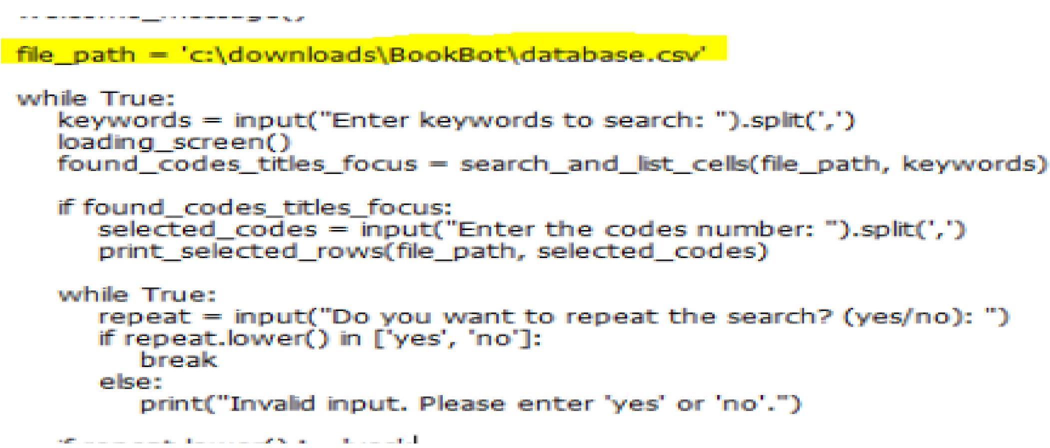


Changing path file:

5. After successfully installing Python, right-click on 'bookbot.py' and choose 'Open with,' then select 'Notepad.'

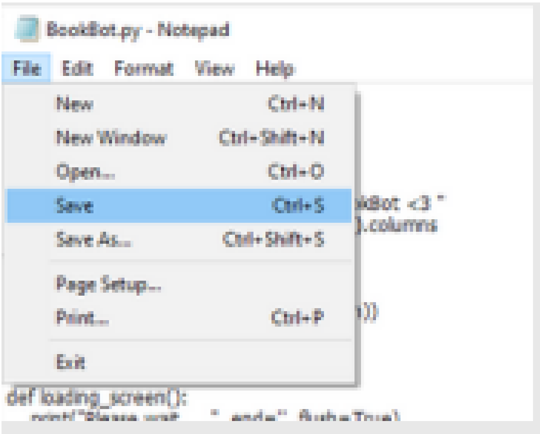


6. Scroll down and locate the line that begins with 'file_path =' in the code. Change the location to your desired database list location.

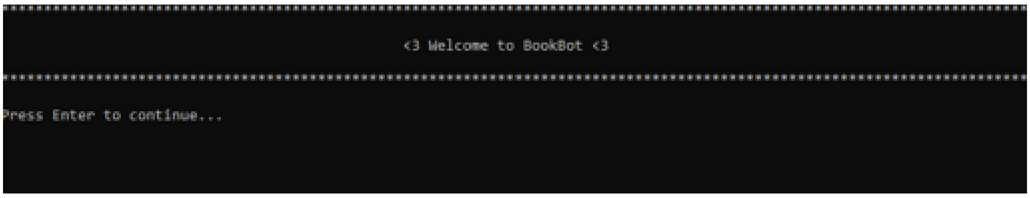


```
file_path = 'c:\downloads\BookBot\database.csv'

while True:
    keywords = input("Enter keywords to search: ").split(',')
    loading_screen()
    found_codes_titles_focus = search_and_list_cells(file_path, keywords)

    if found_codes_titles_focus:
        selected_codes = input("Enter the codes number: ").split(',')
        print_selected_rows(file_path, selected_codes)

    while True:
        repeat = input("Do you want to repeat the search? (yes/no): ")
        if repeat.lower() in ['yes', 'no']:
            break
        else:
            print("Invalid input. Please enter 'yes' or 'no'.")
```

7. save the notepad

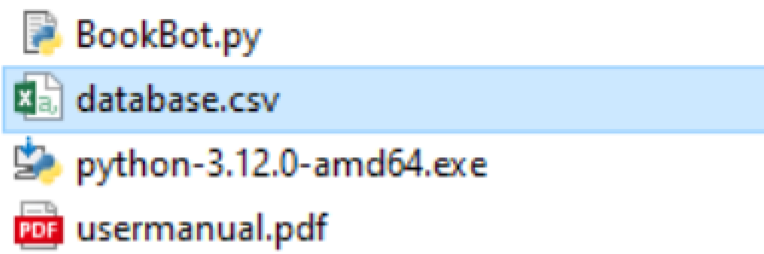


8. and open the “bookbot.py”



How to edit and input a new thesis paper:

1. open ‘database.csv’



2. edit or add new thesis paper

Code	Title	Author	Focus	Year	Location	Call Number
1	Online Consumer's Purchasing Intention: The Influenc	Trisha Fe A. Giray Rayman T. Quinanes Leslie G. Sibayan Rosalie C. Valencia	Statistics	2022	Theses	T G523 2022 Stat.
2	Student's Experience in Online Learning: Preferred M	Hannah Alizza A. Banaag Abegail M. Marababon Mary Louise A. Montoper Jay Nomer C. Radana	Statistics	2022	Theses	T B212 2022 Stat.
3	Consumer Adaption of Online Food Delivery Services	Marloun N. Condes Daniella S. Garing Jakylene M. Perez Abegail S. Salipot	Statistics	2022	Theses	T C745 2022 Stat.
4	The Impact of Implementing the New Normal to Rizal T	Dynzel Jeun L. Canuto Krizia E. Macasero Megan S. Mara?a Angelica Love Uzziel Semilla	Statistics	2022	Theses	T C235 2022 Stat.
5	Mobile Food Ordering Apps: Consumer's E-Satisfacti	Micaella Louis S. Bojo Alyssa Marie C. Gabinete Diane Geoglu B. Mercado Ray Allen M. Reyes	Statistics	2022	Theses	T B685 2022 Stat.
6	Filipino Consumer's Online Shopping Behavior	Genesis E. Cavillo Russell L. De Jesus James Nicole Merilles Joanna Marie A. Urbano	Statistics	2022	Theses	T C419 2022 Stat.
7	Employment Rate, Education, and Population as Dete	Jean Ara R. Cantoria Chloe Marie C. Cura Isha Marie P. Mabuhay Kimberly B. Rayanon	Statistics	2022	Theses	T C232 2022 Stat.
8	Perspective of Rizal Technological University Studen	Kenzein M. Barrientos Nika May R. Carsula Mica Camille E. Diaz Jingky A. Dorota	Statistics	2022	Theses	T B275 2022 Stat.
		Sherly P. Arcilla				T

**Please enter the data into the appropriate cells, such as code, title, year, author, focus, location, and call number.

3. don't forget to save!

Developer Organization

Personnel	Roles	Day/s Active in Meeting
Candelaria, Arvin H. 	Definition of Terms System Available in Global Market	**1st Meeting:** November 26, 2023 **2nd Meeting:** November 29, 2023 (Face-to-Face Meeting) **3rd Meeting** December 5, 2023 **4th Meeting:** December 8, 2023
Dela Cruz, Jersey Jade G. 	System available in Global Market Encoder(User Manual)	**1st Meeting:** November 26, 2023 **2nd Meeting:** November 29, 2023 (Face-to-Face Meeting) **3rd Meeting** December 5, 2023 **4th Meeting:** December 8, 2023
Francisco, James S. 	Encoder(User Manual) Enhancement Scope	**1st Meeting:** November 26, 2023 **2nd Meeting:** November 29, 2023 (Face-to-Face Meeting) **3rd Meeting** December 5, 2023 **4th Meeting:** December 8,

		2023
Rodriquez, Joshua B. 	Programmer Encoder(User Manual, Source Code)	**1st Meeting:** November 26, 2023 **2nd Meeting:** November 29, 2023 (Face-to-Face Meeting) **3rd Meeting** December 5, 2023 **4th Meeting:** December 8, 2023
Sotelo, Chris Joshua P. 	Encoder(User Manual, Bibliography) Description of the Proposed System Developer Organization	**1st Meeting:** November 26, 2023 **2nd Meeting:** November 29, 2023 (Face-to-Face Meeting) **3rd Meeting** December 5, 2023 **4th Meeting:** December 8, 2023
Uy, Francis Oliver P. 	Copy Editor/Copyreader Encoder(User Manual, Bibliography) Description of the Proposed System Purpose	**1st Meeting:** November 26, 2023 **2nd Meeting:** November 29, 2023 (Face-to-Face Meeting) **3rd Meeting** December 5, 2023 **4th Meeting:** December 8, 2023

Source Code

```
import csv
import time
import shutil


def welcome_message():
    welcome_text = "<3 Welcome to BookBot <3 "
    line_length = shutil.get_terminal_size().columns
    line = '*' * line_length


    print(line.center(line_length))
    print(welcome_text.center(line_length))
    print(line.center(line_length))
    input("Press Enter to continue...")


def loading_screen():
    print("Please wait.....", end='', flush=True)
    for _ in range(10):
        print(".", end='', flush=True)
        time.sleep(0.2)
    print()


def search_and_list_cells(file_path, keywords):
    found_codes_titles_focus = []


    while True:
```

```
        with open(file_path, 'r') as csvfile:
            csv_reader = csv.DictReader(csvfile)

            found_cells = []

            for row in csv_reader:
                for keyword in keywords:
                    for column, value in row.items():
                        if keyword.lower() in value.lower():
                                   found_cells.append((column,
value))

found_codes_titles_focus.append((row['Code'],     row['Focus'],
row['Title']))

            if not found_cells:
                print("Keywords not found! Please try again.")
                   keywords = input("Enter keywords to search:
").split(',')
            else:
                print("Keywords found:")
                                               line_length   =
shutil.get_terminal_size().columns
                print("-" * line_length)
                   print("{:<15} {:<30} {:<15}".format("Code",
"Focus", "Title"))
```

```
                print("-" * line_length)
                                for code, title, focus in
found_codes_titles_focus:
                                        print("{:<15} {:<30}
{:<15}\n".format(code, title, focus))
                break

    return found_codes_titles_focus

def print_selected_rows(file_path, selected_codes):
    with open(file_path, 'r') as csvfile:
        csv_reader = csv.DictReader(csvfile)

        selected_codes_set = set(selected_codes)

        found = False

        for row in csv_reader:
            if row['Code'] in selected_codes_set:
                found = True
                                              line_length =
shutil.get_terminal_size().columns
                print(f"*" * line_length)
                print(f"\nThesis Code # {row['Code']}:\n")
                print(f"Title: {row['Title']}")
                print(f"Author:\n{row['Author']}")
```

```
                print(f"Focus: {row['Focus']}")
                print(f"Date: {row['Year']}")
                print(f"Location: {row['Location']} Corner")
                print(f"Call Number:\n{row['Call Number']}\n")
                print(f"*" * line_length)

        if not found:
            print("Thesis code incorrect. Please enter a valid
code.")



def main():
    welcome_message()

    file_path = 'D:\DOWNLOADS\BookBot\prog\database.csv'

    while True:
        keywords = input("Enter keywords to search:
").split(',')
        loading_screen()
        found_codes_titles_focus =
search_and_list_cells(file_path, keywords)

        if found_codes_titles_focus:
            selected_codes = input("Enter the codes number:
").split(',')
```

```
            print_selected_rows(file_path, selected_codes)

        while True:
            repeat = input("Do you want to repeat the search? (yes/no): ")
            if repeat.lower() in ['yes', 'no']:
                break
            else:
                print("Invalid input. Please enter 'yes' or 'no'.")

        if repeat.lower() != 'yes':
            print("\nThank you for using BookBot. Enjoy reading!")

            credits_text = "BookBot © 2023"
            credits_jb = "Joshua Rodriguez"
            credits_ac = "Arvin Candelaria"
            credits_jd = "Jerseyjade Dela Cruz"
            credits_jf = "James Francisco"
            credits_cs = "Chris Joshua Sotelo"
            credits_ou = "Francis Oliver Uy"
            credits_ar = "All Right Reserved"
            line_length2 = shutil.get_terminal_size().columns

            print(credits_text.center(line_length2))
```

```
            print(credits_jb.center(line_length2))
            print(credits_ac.center(line_length2))
            print(credits_jd.center(line_length2))
            print(credits_jf.center(line_length2))
            print(credits_cs.center(line_length2))
            print(credits_ou.center(line_length2))
            print ("\n")
            print(credits_ar.center(line_length2))
            input("Press Enter to exit")
            break

if __name__ == "__main__":
    main()
```

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